

Shuhao Zhang

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Profile

Shuhao Zhang is a Professor in the School of Computer Science and Technology at Huazhong University of Science and Technology. He leads a task under a major national science and technology project of the Ministry of Science and Technology of China and is a recipient of the National Natural Science Foundation of China Excellent Young Scientists Fund (Overseas). In 2026, he was awarded support through the NSFC Research Fund for International Excellent Young Scientists (RFIS-II).

His research focuses on efficient execution and state management in parallel and distributed systems: hardware-conscious execution, shared-state concurrency control, online migration, and recovery. His current systems agenda centers on inference serving for domestic accelerators in China, using vLLM-HUST to study request scheduling, KV-state management, and efficient execution. Related work explores reasoning workflows, external memory, and dynamic retrieval.

Appointments

Huazhong University of Science and Technology

Professor

School of Computer Science and Technology.

Nanyang Technological University

Assistant Professor

2023-2024.

Singapore University of Technology and Design

Assistant Professor

2021-2023.

Technische Universität Berlin

Postdoctoral Researcher

Postdoctoral research on data systems and distributed stream processing.

Education

National University of Singapore

Ph.D., Computer Science, 2019

Research area: database systems, stream processing, parallel and distributed systems.

Nanyang Technological University

B.Eng., Computer Engineering, 2014

Research Focus

- Hardware-conscious execution on multicore and heterogeneous architectures.
- Shared-state concurrency control, online migration, and failure recovery.
- Cost-aware inference scheduling, reasoning workflows, and state reuse.

Major Projects and Funding

- Principal Investigator, NSFC Research Fund for International Excellent Young Scientists (RFIS-II), awarded in 2026.
- Task Lead, major national science and technology project of the Ministry of Science and Technology of China.
- Principal Investigator, National Natural Science Foundation of China Excellent Young Scientists Fund (Overseas).
- Principal Investigator and collaborator on previous research grants from Singapore MOE, Singapore NRF, Temasek Laboratories, and university-level research programs.

Representative Systems

- **vLLM-HUST / BidKV**: Inference serving on domestic accelerators and utility-guided preemption under KV pressure.
- **SAGE**: Dataflow-native execution for modular and controllable LLM reasoning.
- **Neuromem**: External-memory lifecycle decomposition and evaluation.
- **BriskSeed / FlowRAG / StreamFP / GRACE / CANDOR-Bench**: Online search-history reuse, dynamic retrieval, stream learning, graph maintenance, and continuous-ANNS evaluation.
- **MorphStream / Spacker / CStream**: Dependency-driven scheduling, distributed state migration, and hardware-conscious stream processing.

Selected Recent Publications

- **BriskSeed: Online History-Guided Reuse for Accelerating Dynamic Approximate Nearest Neighbor Search**. Hongru Gao, **Shuhao Zhang**, Haikun Liu, Xiaofei Liao, Hai Jin. ICDE 2027. Accepted. Corresponding Author.
- **BidKV: Utility-Guided Preemption Scheduling for KV-Pressure LLM Serving**. SC 2026. Corresponding Author.
- **SAGE: A Dataflow-Native Framework for Modular, Controllable, and Transparent LLM-Augmented Reasoning**. ICML 2026. Corresponding Author.
- **Neuromem: A Granular Decomposition of the Streaming Lifecycle in External Memory for LLMs**. ICML 2026. Corresponding Author.
- **CANDOR-Bench: Benchmarking In-Memory Continuous ANNS under Dynamic Open-World Streams**. SIGMOD 2026. Corresponding Author.
- **MorphStream: Adaptive Scheduling for Scalable Transactional Stream Processing on Multicores**. SIGMOD 2023 / TKDE 2025 line of work. Corresponding Author.
- **Towards Concurrent Stateful Stream Processing on Multicore Processors**. ICDE 2020. First Author.
- **BriskStream: Scaling Data Stream Processing on Shared-Memory Multicore Architectures**. SIGMOD 2019. First Author.

Honors and Service

- HUST Excellent Young Scholar; Huawei Donghu Young Scholar.
- VLDB Review Board Member; SC Program Committee Member; ICDE 2024 Poster Track Chair.
- Program committee service for KDD, CIKM, DASFAA, SDM, and related venues.

Teaching and Open Materials

- Large Language Model Inference Systems and Practice, a practice-oriented graduate course connected to vLLM-HUST development.
- Graduate Paper Writing, public lecture materials for research writing and systems-paper methodology.